**FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)**

*First Year and Foundation Unit*

*Joint Group Assignment (JGA) 2026 — Theme: Digital Entrepreneurship*

# Student Step-by-Step Guide

## What this guide is for

This guide walks you through the JGA 2026 project from the day your group is formed to the day you submit your final report. It does not replace the official assignment brief, read that first, and keep it open beside this guide. Here you will find the order to do things in, what to produce at each step, and what your lecturer is expecting at each phase.

You are being introduced to the project at the end of the first semester. Active project work runs from 1 June to 30 September, with the winter holiday (1 to 11 July) as a break. You will end the project with three connected deliverables: a live e-commerce or subscription-based website or app, a formal investigative report, and a group presentation. Half of your subject marks come from how well you apply what you have learned in your Mathematics, Communication, Programming and Computing modules — so use this project to practise those skills, not as a separate task.

## The project at a glance

| | |
|---|---|
| **Theme** | Digital Entrepreneurship for Sustainable Community Development. |
| **Group size** | 7 to 10 students from your timetable main group. |
| **Active project window** | 1 June to 30 September.<br>Winter holiday: 1 to 11 July (no project work expected). |
| **Final products** | 1. A hosted e-commerce website OR subscription-based app (prototype).<br>2. A formal investigative report.<br>3. An in-class group presentation. |
| **Weight** | See your Module's Study Guide |

## Your timeline at a glance

The project is split into four phases. Phases 1 to 4 run across the June-to-September window, with the winter holiday between Phases 1 and 2.

| Phase | Approximate window | Main focus |
|---|---|---|
| **Phase 0** | Introduction (end of May) | Form groups, choose an idea, set up how you will work together. |
| **Phase 1** | 1 June – 30 June (about 4 weeks) | Plan and design on paper. No coding yet. |
| **Winter break** | 1 July – 11 July | No project work expected. Rest, then come back ready. |
| **Phase 2** | 14 July – 10 August (about 4 weeks) | Build the first working Java version. Start the formal report. |
| **Phase 3** | 11 August – 7 September (about 4 weeks) | Build the website, host it online, analyse data, draft most of the report. |
| **Phase 4** | 8 September – 30 September (about 3 weeks) | Final testing, recommendations, presentation rehearsal, submission. |

*Your lecturer will confirm exact in-class presentation dates and the final submission date. Use the windows above to plan your weekly work; do not wait for fixed dates to start.*

# Phase 0 — Before 1 June: set the group up properly

**When:** End of May, after you are introduced to the project in class.

**Goal:** Have your group, your shared workspace, and a draft idea ready so that work can begin on 1 June.

Do these four things in your first project meeting. They sound small, but skipping them is where most groups lose marks later.

## Step 1 — Form your group properly

- Form a sub-group of 7 to 10 students from your timetable main group. Your lecturer must approve the group.
- Choose a Project Leader. This person is the contact point with your lecturer and runs your meetings — they are not the one who does all the work.
- Choose a Scribe to take minutes at every meeting. Keep all minutes in one shared folder; you will reference them in your final report.

## Step 2 — Set up how you will work together

- Set up a cloud-based collaborative space (Microsoft Teams, OneDrive, Google Drive or GitHub) where everyone can access shared files.
- Agree on one channel for fast communication (e.g. WhatsApp) and one for formal communication (email or Teams).
- Agree on a weekly meeting time. Put it on a shared calendar.

## Step 3 — Choose your business idea

Pick an idea that solves a real problem you can describe in one sentence. The brief lists examples: online tutoring, food ordering, clothing stores, student marketplaces — but you are free to propose your own. Use the linked resources in the brief for inspiration.

Test the idea against three quick questions before you commit:

1. Who exactly is the customer? (You must be able to name a real person who would use this.)
2. What problem are you solving for them? (One sentence.)
3. How will the business make money? (Ads, subscriptions, e-commerce, premium — pick one and explain why.)

## Step 4 — Submit your group list

- Submit your full group list (names, surnames, student numbers, and the Project Leader marked) to your lecturer before 1 June.
- Once submitted, the group is locked. Members cannot be swapped without your lecturer's approval.

# Phase 1 — 1 June to 30 June: plan and design (no coding yet)

**Duration:** About 4 weeks.

**Goal:** Understand the problem, design the solution on paper, and prove your group can work together. You should not write any Java code in this phase.

## Step 5 — Define your problem domain

- Write a one-paragraph problem statement: who has the problem, what the problem is, and why it matters.
- Identify the variables and parameters your system needs. A variable is something that changes (e.g. number of orders today); a parameter is something fixed for now (e.g. delivery fee). List them in a simple two-column table.
- Do basic financial computations: estimate cash flow, expected profits, possible losses, and a simple cost-benefit analysis for the first year of your business. Use Microsoft Excel for this and keep the file.

## Step 6 — Design the solution on paper

- Draw a flowchart of how a user moves through your website or app, from landing page to a completed purchase or sign-up.
- Write pseudocode for your main feature. Pseudocode is plain-English steps, numbered, that a programmer could turn into code. It is not Java.
- Build an IPO (Input – Processing – Output) table for each menu option in your prototype. For every input, list what could go wrong (an invalid entry) and how the program should respond.

## Step 7 — Research the field

- Find at least five (5) real-world digital businesses similar to your idea. Take screenshots and note what they do well and what they do badly.
- Save every source as you find it — URL, author, date accessed. Do this now; do not try to reconstruct it later.
- Demonstrate proficiency in internet search: use specific search terms, not single words. For example, search "South African student food delivery startup" rather than "food app".

## Step 8 — Prepare your Phase 1 presentation

At the end of Phase 1, each group presents a short summary of where they are. Your presentation must include:

- Group members and roles.
- The problem and the proposed solution (one slide each).
- Your IPO table and flowchart.
- Initial financial computations.
- A summary of the five businesses you researched.

### Phase 1 checklist

- Group list with Project Leader, submitted.
- Shared cloud workspace set up.
- Problem statement (one paragraph).
- Variables and parameters table.
- Cost-benefit Excel spreadsheet.
- Pseudocode and flowchart of the main feature.
- IPO table covering all menu options and error cases.
- Five researched case studies with screenshots and sources.
- Phase 1 PowerPoint presentation.

## Winter break — 1 July to 11 July

No project work is expected during the winter holiday. Rest. If you want to use the time productively, three small things will pay off later without costing you the break:

- Glance at one or two of the recommended website builders (WordPress, Wix or Google Sites) so you walk into Phase 2 with a feel for the tool.
- Make sure your shared cloud workspace still works and everyone can still log in.
- Re-read your Phase 1 deliverables once before the break ends so you start Phase 2 with the design fresh in your mind.

# Phase 2 — 14 July to 10 August: first working version

**Duration:** About 4 weeks.

**Goal:** Turn your Phase 1 design into a working first version of the program in Java.

Start writing the formal report.

## Step 9 — Write the first Java version

- Open your Phase 1 pseudocode next to your editor and translate it line by line into Java. Do not skip the pseudocode and code from scratch — you will lose track of the logic.
- Implement the basic menu using if-statements or a switch statement. Each option should call a method, not a long block of inline code.
- Implement the mathematical operations you identified in Phase 1 (totals, percentages, simple statistics).
- Add a comment block at the top of every file (author, date, purpose) and inline comments on any line that is not obvious.
- Use consistent indentation. Java does not require it, but your lecturer does.

## Step 10 — Convert your maths into code

Take the equations you wrote in Phase 1 and turn them into Java statements.

Examples of what should appear in your code:

- Variable declarations with sensible names (totalRevenue, not x).
- Arithmetic operations using the correct data types (use double for money, not int).
- Logical conditions (if the order value exceeds R500, apply free delivery).

## Step 11 — Start the formal report

- Generate a research title that names the problem and the solution. Example shape: "A digital marketplace for [target users] in [context]: A prototype investigation."
- Write the Aim and Objectives. The Aim is one sentence about what the project will achieve overall. The Objectives are 3 to 5 measurable steps that get you there.
- Write a Progress Report (1 to 2 pages) covering what has been researched, designed and coded so far, and what is still outstanding.
- Write a Problem Statement based on a case study — pick one of your five researched businesses and describe the gap your prototype fills compared to it.

## Step 12 — Organise findings and prepare for evaluation

- Move your research notes from Phase 1 into Microsoft Excel: one row per source, with columns for URL, summary, what you used it for, and date accessed.
- Begin a Word document for the report using proper headings, page numbers, and a table of contents. Set Times New Roman or Arial size 12, as required by the brief.
- Make sure every group member can explain their own contribution out loud. Your lecturer may ask any member, not just the Project Leader.

## Phase 2 checklist

- Java program with working menu and basic operations.
- Inline comments and a header comment block in every file.
- Research title.
- Aim and Objectives (in the report).
- Progress report (1 to 2 pages).
- Problem statement based on a case study.
- Excel file of organised research sources.
- Report Word document started, with proper formatting.
- Each group member able to explain their contribution.

# Phase 3 — 11 August to 7 September: build the website and analyse the data

**Duration:** About 4 weeks.

**Goal:** Strengthen the Java program, build the actual website, and turn your raw data into charts and findings.

## Step 13 — Strengthen the Java program

- Add input validation everywhere. If you ask for a number between 1 and 5, the program must reject 0, 7 and the letter 'a' without crashing.
- Replace single if-statements with loops where the user might want to do the same action multiple times (e.g. ordering more than one item).
- Refactor for readability: rename confusing variables, split methods that have grown longer than about 30 lines, and remove any code you are not using.
- Begin basic integration between Java and JavaScript or Web 2.0 content where it makes sense for your prototype.

## Step 14 — Build and host the website

- Choose your platform from the brief: WordPress, Wix, Google Sites, or plain HTML/CSS. Pick what your group can actually maintain — a basic site that works beats a fancy site that breaks.
- Build pages for: Home, About (1 paragraph on what problem you solve and who you solve it for), Customer Persona (a card or infographic for your target user), Products or Services, Revenue Model, and Contact.
- Embed or link the materials you researched. When you use someone else's image, video or text, reference the source on the same page.
- Get the site hosted online. A site that only runs on your laptop is worth 0 of the 10% allocated to a hosted, running platform.

## Step 15 — Apply mathematics to your data

- Apply measurements and conversions to the variables you defined in Phase 1 — for example, convert daily orders into monthly figures, or rand-per-customer into rand-per-month.
- Use elementary statistics (mean, median, range, percentages) to summarise the effects of your parameters.
- Build charts and graphs in Excel: at least one chart per major variable. Choose the right chart type — bar charts for comparing categories, line charts for trends over time, pie charts only for parts of a whole.
- Export the charts as images and add them to both your website and your report.

## Step 16 — Strengthen the report

- Write the Introduction: brief background to the problem, why it matters in the South African context, and what your project sets out to do.
- List your sources properly. Use Chicago Author-Date or another consistent style and apply it everywhere — do not mix styles.
- Add the Presentation of Findings section: summarise what your five researched businesses do, what you learned, and how it shaped your prototype.
- Write a Conclusion that links findings back to your Aim and Objectives.

## Phase 3 checklist

- Java program with validation, loops, and clean structure.
- Live, hosted website with all required pages.
- Sources referenced on the website.
- Excel charts and graphs.
- Report sections: Introduction, Findings, Sources, Conclusion.
- Updated PowerPoint for the Phase 3 presentation.

# Phase 4 — 8 September to 30 September: final polish and submission

**Duration:** About 3 weeks.

**Goal:** Finalise everything, fix every bug you can find, and submit a clean package before 30 September.

## Step 17 — Final Java testing

- Run the program through every menu option in order. Then run it again, entering deliberately bad input at every prompt (letters where numbers go, negative numbers, very large numbers). Fix anything that crashes or misbehaves.
- Remove every block of code you are not using. Dead code costs you marks.
- Check that your validation messages are clear and polite ("Please enter a number between 1 and 5", not "ERROR").

## Step 18 — Cybersecurity and cloud basics

- Apply basic cybersecurity principles to the website: do not store passwords in plain text, do not put real personal data in a demo, and use HTTPS if your host offers it.
- Document in the report what hosting environment you used, what database (if any), and what development tools. One paragraph is enough.

## Step 19 — Write the final recommendations

- Recommendations should answer one question: How could this kind of digital business foster innovation and job creation in South Africa?
- Make recommendations specific. "The government should support entrepreneurs" is too vague. "A six-month mentorship and free hosting credits for first-year ICT students would lower the barrier to launching a prototype" is specific.
- Recommend two or three integrations that would extend your prototype — for example, an AI chatbot for customer queries, a no-code workflow to automate order confirmations, or a payment gateway for South African users.

## Step 20 — Compile the final report

Your final report should contain, in this order:

4. Cover page (initials, surnames, student numbers, signatures of all 7 to 10 members; subject name and code; qualification code and class group; topic; lecturer's surname; submission date).
5. Table of contents.
6. Aim and Objectives.
7. Problem Statement.
8. Introduction.
9. Findings (your five case studies and your prototype description).
10. Methods and Tools used (programming language, hosting, AI or no-code tools).
11. Results (charts, summaries, analytics).
12. Recommendations.

13. Conclusion.
14. Reference list.
15. Appendices (Java source code, pseudocode, screenshots of the live site, evidence of teamwork such as meeting minutes).
16. Rubric (attached as required by the brief).

## Step 21 — Prepare the final presentation

- Plan a 2-minute elevator pitch: Problem → Solution → Who it is for → How AI makes it better.
- Decide who presents what. Every group member should speak — marks are partly for teamwork visible during the presentation.
- Have the live website open in a browser tab, ready to demonstrate. Do not rely on screenshots if the site is supposed to be live.
- Prepare for questions. Your lecturer can ask any member about any part.

## Step 22 — Submit

- Submit to your lecturer in person before 30 September. Emailed submissions will not be accepted — re-read this clause in the brief before submission day.
- Bring: printed report (or as your lecturer specifies), the attached rubric and cover page, and the live website URL written on the cover page.
- Aim to finish 3 to 5 days before 30 September. Things go wrong on the last day — printers jam, sites go offline, group members get sick.

## How you will be graded

Half your subject mark comes from this project. The marks are split across eight rubric items, each scored from 1 (Poor) to 6 (Excellent). Aim for a 5 or 6 on every item by attending to the matching column below.

| Rubric item | Weight | What gets you full marks |
|---|---|---|
| Application of Programming | 15% | Working Java code with validation, loops, clean structure, meaningful variable names, and consistent indentation. |
| Application of Mathematics | 15% | Identified variables and parameters used correctly in financial calculations and elementary statistics, with charts that interpret the numbers. |
| Application of Computing | 10% | Demonstrated competence in Word, Excel, PowerPoint, internet search, and cloud collaboration tools across the project. |
| Application of Communication | 10% | Clear academic writing, correct tense and structure, edited language, properly cited sources. |
| Prototype Design & Content | 10% | Website pages cover Business Concept, Customer Persona, Revenue Model, and a clear elevator pitch. Includes AI or no-code tools. |
| Hosted platform running online | 10% | The website is live on a public URL on the day of presentation and works without errors. |
| Report Clarity & Organisation | 15% | Report follows the required structure, uses Times New Roman or Arial size 12, has a table of contents, page numbers, and a complete reference list. |
| Presentation & Teamwork | 15% | Every member speaks, the live demo runs, and questions to any member are answered confidently. |

## Common mistakes to avoid

- Leaving the website until Phase 4. Hosting always takes longer than expected — start in Phase 3.
- Writing code without pseudocode. You will end up debugging logic and syntax at the same time, which doubles the work.
- Treating the winter break as a place to catch up. Use Phase 1 to finish Phase 1 work, not the holiday.
- Copying text from websites without referencing. This is plagiarism and will cost you marks at best, the project at worst.
- One person doing all the work. Marks are split across rubric items partly to reward teamwork — your lecturer can tell when one voice has written everything.
- Submitting the rubric and cover page late or missing. They are required attachments.
- Demonstrating from screenshots when a live site was required. The 10% for a hosted, running platform depends on the site actually running.

## Where to find the resources you need

Your brief lists clickable links. Use them — they are chosen for the assignment.

- Project inspiration: ProjectPro's 50+ Solved AI Projects, and GitHub's First Year AI Projects topic.
- Business ideas: BestProjectIdeas' 300 Student Business Ideas, and PrometAI's Student Startup Examples 2026.
- Website building workshop: UCL's Build & Launch a Website step-by-step guide.
- Recommended tools: WordPress, Wix or Google Sites for the site; Java and JavaScript for code; AWS, Microsoft Azure or GitHub for hosting; MS Office Suite for documents.

## Final advice

Start on 1 June. The students who do well on this project are the ones who treat the Phase 1 deliverables as a real submission, not as a warm-up. By the time you get to Phase 4 you should be polishing, not building.

Keep a single shared folder for the whole project — your pseudocode, your Java files, your Excel sheets, your meeting minutes, your draft report, and your screenshots. When you compile the final submission, everything you need will already be in one place.

Ask your lecturer when you are stuck. The project is designed to be done with weekly guidance, not in isolation. Good luck.